

Coastal Hydrodynamics I

Course : CE60222

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Introduction

- Coastal hydrodynamics outlines the importance of understanding nearshore hydrodynamics for sediment transport.
- It highlights the following aspects:
 1. **Mean and Oscillatory Water Levels:** Discuss how waves, wind, and tides influence water levels and currents.
 2. **Wave Propagation:** Linear wave propagation effects, including how waves change as they approach the shore, such as increasing in height and decreasing in wavelength.
 3. **Non-linear Wave Transformation:** The transformation of wave shapes from symmetric to asymmetric profiles as they near breaking.
 4. **Wave Boundary Layer:** The effects of wave dissipation in the boundary layer on wave velocities and sediment transport will be explored.
 5. **Wave-Induced Water Level Changes:** It will address phenomena like wave set-up (the rise in water level at the coast) and the resulting currents, including cross-shore and longshore currents.

Wave Deformation

Energy balance

- When waves propagate **from deep into intermediate and shallow water**, the waves transform, i.e. **wave height, length and direction change** until the waves finally break and lose their energy.
- **Wave transformation** takes place because the waves are affected by the seabed through processes such as **refraction, shoaling, bottom friction and wave breaking**.
- Numerically (or analytically) solving the **energy balance** yields information on the **wave transformation** of a wave field, while the waves approach the shore.
- In the **presence of a current**, energy is **not conserved** any longer, since transfer of energy between waves and currents is possible.
- In that case another wave quantity, wave action E/ω , will be conserved and the **wave action balance** rather than the **energy balance** should be solved.
- In the **absence of a current**, the **wave action balance** reduces to the **energy balance**.

- Integrating over all frequencies and directions in an irregular wave field, Energy conservation can be composed :

$$\underbrace{\frac{\partial E}{\partial t}}_{\text{change of energy}} + \underbrace{\frac{\partial}{\partial x}(Ec_g \cos \theta)}_{\text{import of energy in } x\text{-direction}} + \underbrace{\frac{\partial}{\partial y}(Ec_g \sin \theta)}_{\text{and in } y\text{-direction}} = \underbrace{S - D}_{\text{gain of energy}} \quad (1)$$

- Definition of wave angles for a wave propagating along a wave ray s is shown in Fig 1.

where

θ = the wave direction angle with respect to the x -axis

S = generation term

D = Dissipation term

C_g = wave group speed

φ = the angle of incidence with respect to the depth contours.

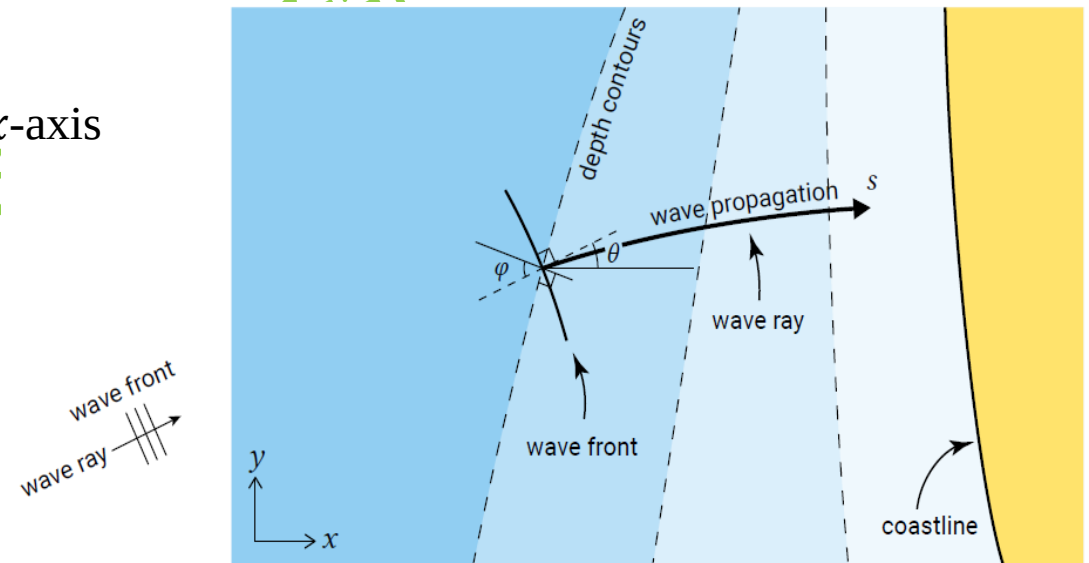


Fig 1. Sketch showing a wave propagating along a wave ray

- If we assume that the wave conditions are stationary (do not change in time). The term $\partial E / \partial t$ on the left-hand side **equals zero**.
- The **energy balance** in the coastal zone can be written as:

$$\frac{d}{dx} (E c_g \cos \theta) + \frac{d}{dy} (E c_g \sin \theta) = -D_f - D_w \quad (2)$$

Where

$E = \frac{1}{8} \rho g H_{rms}^2$ total wave energy being propagated at the wave group speed c_g in the wave propagation direction θ .

D_w = **Wave dissipation** due to wave-breaking

D_f = **Wave dissipation** due to bottom friction

- The **direction** θ can change and we therefore need information on θ to find a solution.
- For simple cases (alongshore uniform coast), **Snell's law** gives this information.

- **Equation 2** can also be written along a wave ray s :

$$\frac{d}{ds}(Ec_g) = -D_f - D_w \quad (3)$$

- Overall, the energy balance is crucial for understanding how waves interact with the coastal environment and the resulting implications for sediment transport and coastal management.

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Shoaling

- **Wave shoaling** is a phenomenon that occurs as ocean waves travel from deeper water into shallower areas, causing changes in **wave height, wavelength, and wave speed**.
- As waves enter shallower water, their **speed decreases**, leading to a **reduction in wavelength** and an **increase in wave height**.

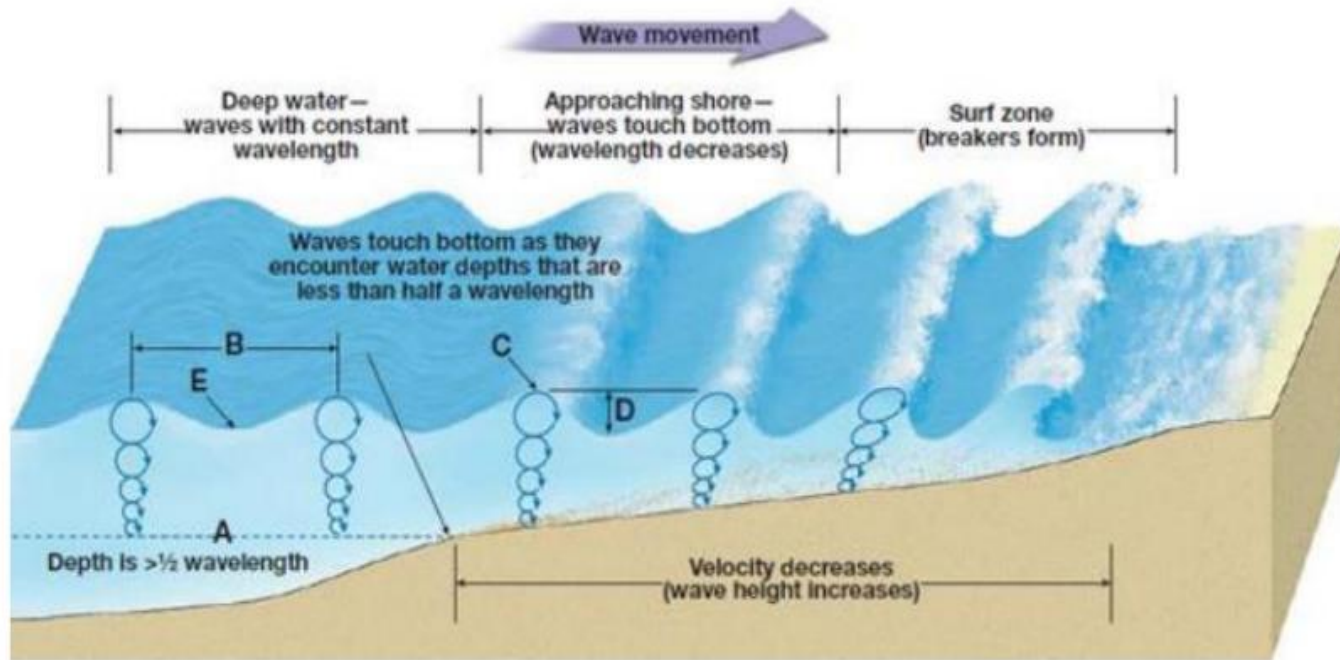


Fig 2. Wave propagation onto the shore

- Outside the **breaker zone** the **dissipation is approximately zero** (neglecting bottom friction) and **integration of Equation-3 yields:**

$$U = Ec_g = Enc = \text{constant} \quad (4)$$

Where, U = wave power or energy flux per unit wave crest width J/(m s)

E = wave energy per unit surface area J/m²

c_g = wave group velocity m/s

c = wave celerity m/s

n = ratio c_g to c

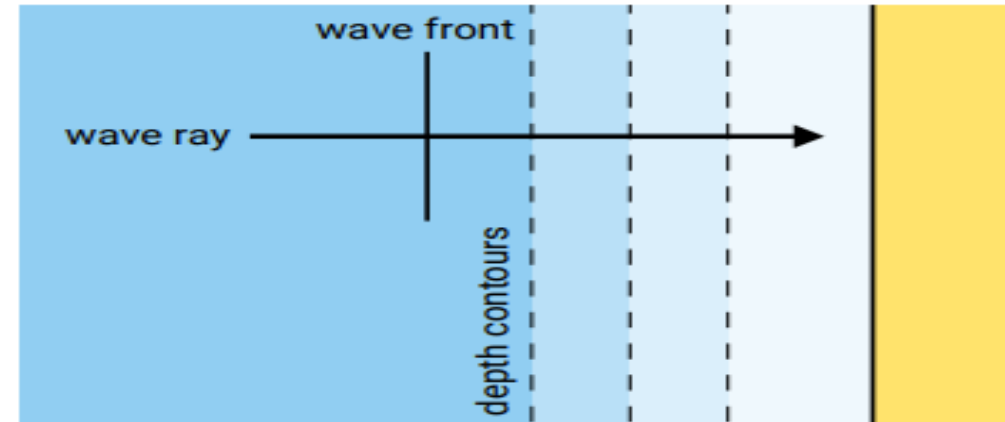


Fig 3. Normally incident waves with parallel depth contours ($\phi = \theta = \text{constant} = 0$).

- **Wave Power:** The **wave energy flux** is the rate at which energy is transmitted in the direction of wave propagation across a vertical plane perpendicular to the direction of wave advance and extending down the depth.

The **average flux per unit** wave crest is given as,

$$P = EnC = EC_g$$

$$\text{where, } n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right]$$

$$\text{Deep water} \quad n=0.5, \quad C_g=0.5C$$

$$\text{Shallow water} \quad n=1, \quad C_g=C$$

$$\text{Transition water} \quad 0.5 < n < 1$$

Deep water conditions

$$\frac{2kd}{\sinh 2kd} = 0 \text{ and } C_g = \frac{1}{2}C \text{ and } n = \frac{1}{2}$$

$$P_0 = \frac{1}{2} E_0 C_0$$

Shallow water conditions

$$\sinh 2kd = 2kd \text{ and } C_g = nC \text{ and } n=1$$

$$P = EC$$

- Assuming the wave propagates from deep water towards the shore and the **ocean bottom slope is gradual** and there are **no undulations** and has **parallel bottom slope contours**.
- Accordingly to the **conservation of energy** Equating the power in the deep waters to that in shallow waters.

$$P_0 = P$$

$$\frac{1}{2} E_0 C_0 = E n C$$

$$\frac{\rho g}{8} H_0^2 \cdot \frac{C_0}{2} = \frac{\rho g}{8} H^2 n C$$

$$\frac{H^2}{H_0^2} = \frac{1}{2n} \cdot \frac{C_0}{C}$$

$$\boxed{\frac{H}{H_0} = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{C}} = K_s}$$

→ Shoaling co-efficient

- The parameter K_s is called the **shoaling factor**.
- In Fig. 4 it is shown as a function of $\frac{H}{L_0}$.
- K_s is 1.0 in deep water, then **decreases slightly with water depth** to 0.91 and **subsequently rises to infinity**.

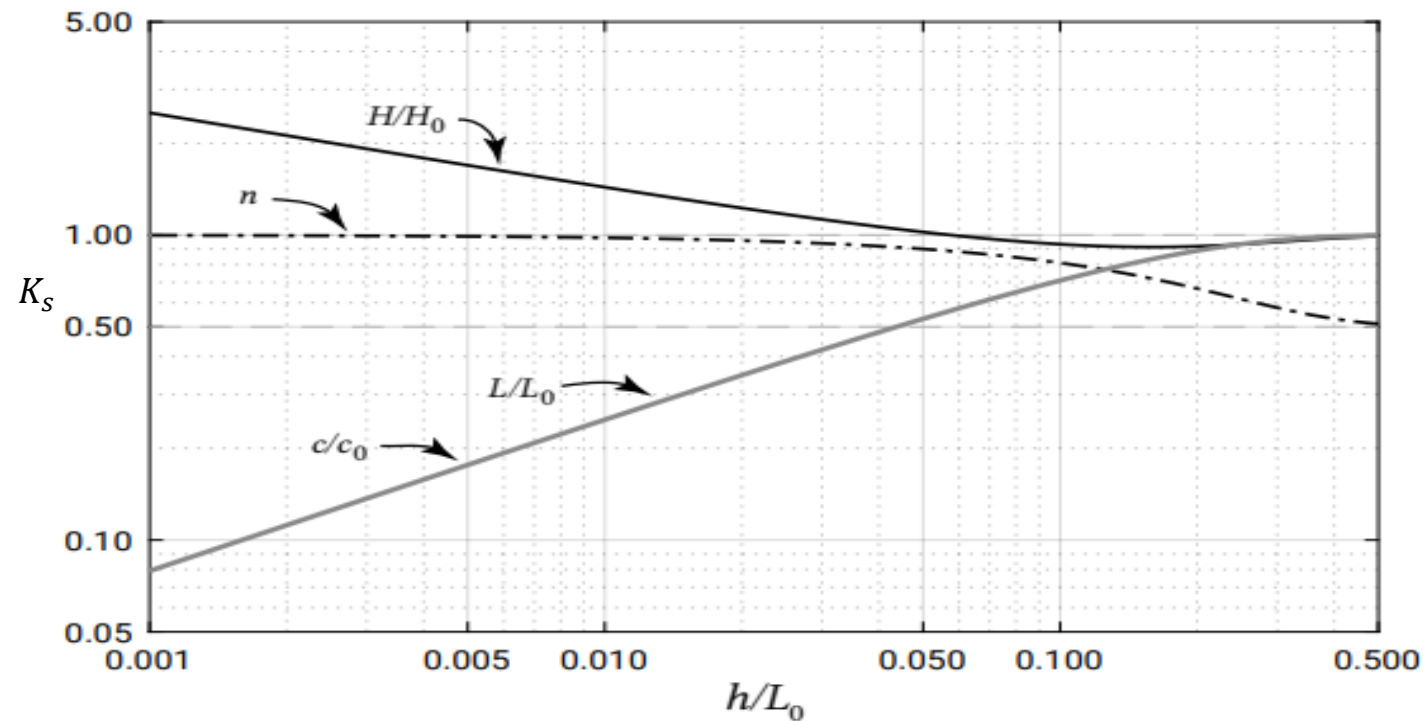


Fig 4. The shoaling factor $K_s = \frac{H}{H_0}$, n and $\frac{C}{C_0} = \frac{L}{L_0}$ as a function of $\frac{h}{L_0}$.

Problem

A 10 s period wave having a height of 5 m propagates from deep to shallow waters. Assuming that the bottom contours are parallel to each other, compute the wave height at a water depth of 10 m.

Solution- Given data, $T = 10$ sec , $d = 10$ m , $H_0 = 5$ m

$$L_0 = 1.56T^2 = 1.56 \times 10^2 = 156 \text{ m}$$

$$\frac{d}{L_0} = \frac{10}{156} = 0.0642$$

$$\frac{d}{L} = 0.1082 \text{ (from wave table)}$$

$$L = \frac{d}{0.1082} = \frac{10}{0.1082} = 92.4 \text{ m}$$

$$C_0 = \frac{L_0}{T} = \frac{156}{10} = 15.6 \text{ m/s}$$

$$C = \frac{L}{T} = \frac{92.4}{10} = 9.24 \text{ m/s}$$

d/L_0	d/L	$2\pi d/L$	$\tanh(2\pi d/L)$	$\sinh(2\pi d/L)$	$\cosh(2\pi d/L)$	H/H_0	K	$4\pi d/L$	$\sinh(4\pi d/L)$	$\cosh(4\pi d/L)$	n	C_0/C_0	M
0.0510	0.0952	0.5981	0.5357	0.6345	1.1843	1.0195	0.8444	1.1963	1.5028	1.8051	0.8980	0.4811	17.19
0.0520	0.0962	0.6047	0.5404	0.6422	1.1884	1.0162	0.8414	1.2093	1.5264	1.8248	0.8961	0.4842	16.90
0.0530	0.0973	0.6111	0.5449	0.6499	1.1926	1.0130	0.8385	1.2222	1.5501	1.8447	0.8942	0.4873	16.62
0.0540	0.0983	0.6175	0.5494	0.6576	1.1968	1.0099	0.8355	1.2351	1.5740	1.8648	0.8924	0.4903	16.35
0.0550	0.0993	0.6239	0.5539	0.6652	1.2011	1.0069	0.8326	1.2479	1.5979	1.8850	0.8905	0.4932	16.09
0.0560	0.1003	0.6303	0.5583	0.6729	1.2053	1.0040	0.8297	1.2606	1.6220	1.9055	0.8886	0.4961	15.83
0.0570	0.1013	0.6366	0.5626	0.6805	1.2096	1.0012	0.8267	1.2732	1.6462	1.9261	0.8867	0.4989	15.59
0.0580	0.1023	0.6429	0.5669	0.6881	1.2139	0.9984	0.8238	1.2858	1.6705	1.9469	0.8848	0.5016	15.36
0.0590	0.1033	0.6491	0.5711	0.6957	1.2182	0.9958	0.8209	1.2983	1.6950	1.9680	0.8830	0.5043	15.13
0.0600	0.1043	0.6553	0.5753	0.7033	1.2225	0.9932	0.8180	1.3107	1.7196	1.9892	0.8811	0.5069	14.91
0.0610	0.1053	0.6615	0.5794	0.7108	1.2269	0.9907	0.8151	1.3231	1.7443	2.0106	0.8793	0.5094	14.70
0.0620	0.1063	0.6677	0.5835	0.7184	1.2313	0.9883	0.8121	1.3354	1.7691	2.0322	0.8774	0.5119	14.50
0.0630	0.1072	0.6738	0.5875	0.7260	1.2357	0.9859	0.8092	1.3476	1.7942	2.0540	0.8756	0.5144	14.30
0.0640	0.1082	0.6799	0.5915	0.7335	1.2402	0.9837	0.8063	1.3598	1.8193	2.0760	0.8737	0.5168	14.11
0.0650	0.1092	0.6860	0.5954	0.7410	1.2446	0.9814	0.8034	1.3719	1.8447	2.0983	0.8719	0.5191	13.92
0.0660	0.1101	0.6920	0.5993	0.7486	1.2491	0.9793	0.8006	1.3840	1.8701	2.1207	0.8700	0.5214	13.74
0.0670	0.1111	0.6980	0.6031	0.7561	1.2537	0.9772	0.7977	1.3960	1.8958	2.1433	0.8682	0.5236	13.57
0.0680	0.1120	0.7040	0.6069	0.7636	1.2582	0.9752	0.7948	1.4080	1.9216	2.1662	0.8664	0.5258	13.40
0.0690	0.1130	0.7100	0.6107	0.7711	1.2628	0.9732	0.7919	1.4199	1.9475	2.1893	0.8645	0.5279	13.23
0.0700	0.1139	0.7159	0.6144	0.7786	1.2674	0.9713	0.7890	1.4318	1.9737	2.2126	0.8627	0.5300	13.07

$$k = \frac{2\pi}{L} = 0.06523$$

$$n = \frac{1}{2} \left(1 + \frac{2kd}{\sinh 2kd} \right) = 0.88 \quad (\text{as } 2kd = 1.31)$$

$$\frac{H}{H_0} = \sqrt{\frac{C_0}{C} \cdot \frac{1}{2n}} = 0.98$$

$$\frac{H}{5} = 0.98$$

$$H = 0.98 \times 5 = 4.91\text{m}$$

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Refraction

- Instead of a normally incident wave, consider now an **obliquely incident linear wave** approaching at a deep-water **angle φ_0** to the shore.
- The wave is again long-crested and the bottom contours are essentially straight and parallel as shown in Fig.4.

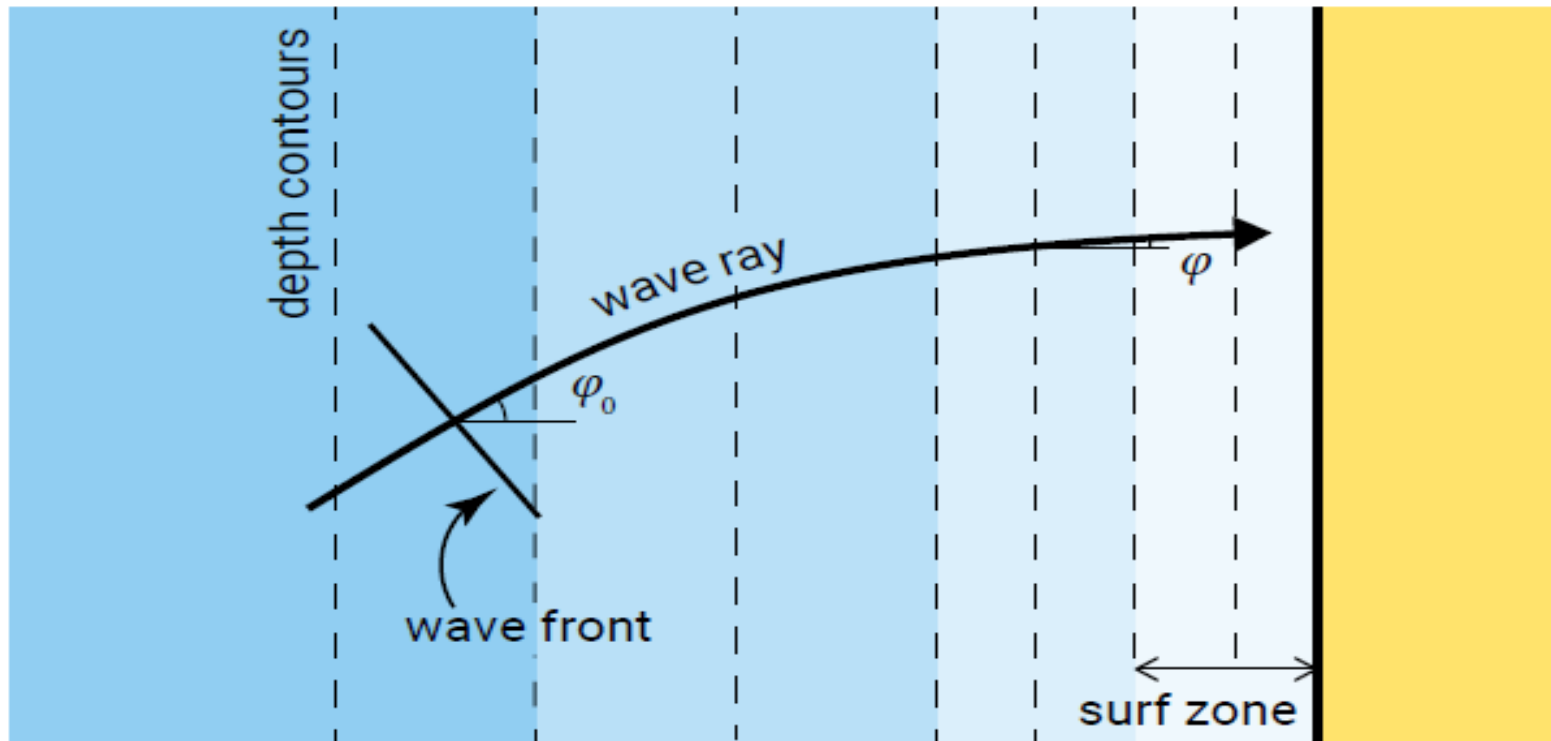


Fig 5. Obliquely incident wave propagating on alongshore uniform depth contours

- When a wave approaches underwater contours at an angle, it is evident that the sections of the crest in the deeper parts travel faster than those in the shallower sectors.
- This causes the wave crest to turn towards the depth contour. This **bending effect** is called **Refraction**.

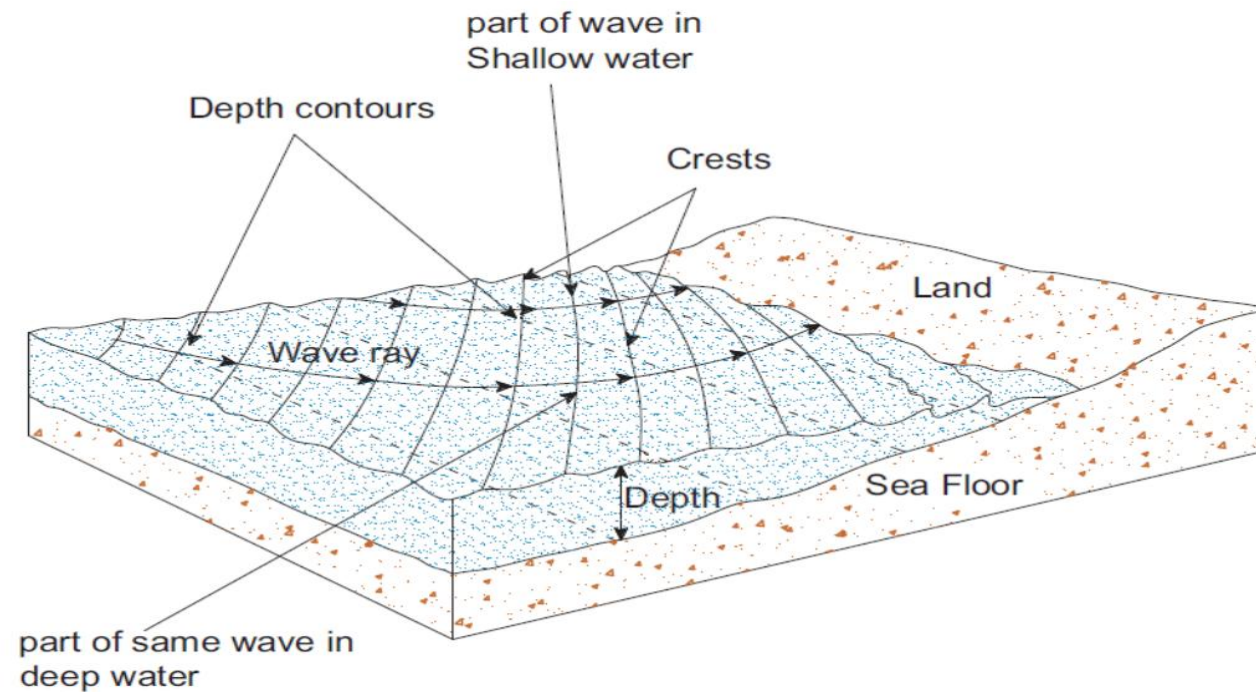


Fig 6. Slowing and bending of waves as they approach shore

- **Construction of wave refraction diagrams**

A **train of waves** travels over a step where the depth instantaneously decreases from d_1 to d_2 , causing the wave celerity C_1 and L_1 to C_2 and L_2 respectively. From figure 7, we can observe that:

$$\sin \alpha_1 = \frac{L_1}{x} = \frac{C_1 T}{x}$$

$$\sin \alpha_2 = \frac{L_2}{x} = \frac{C_2 T}{x}$$

Dividing:

$$\frac{\sin \alpha_1}{\sin \alpha_2} = \frac{C_1}{C_2} = \frac{L_1}{L_2}$$

→ **Snell's Law**

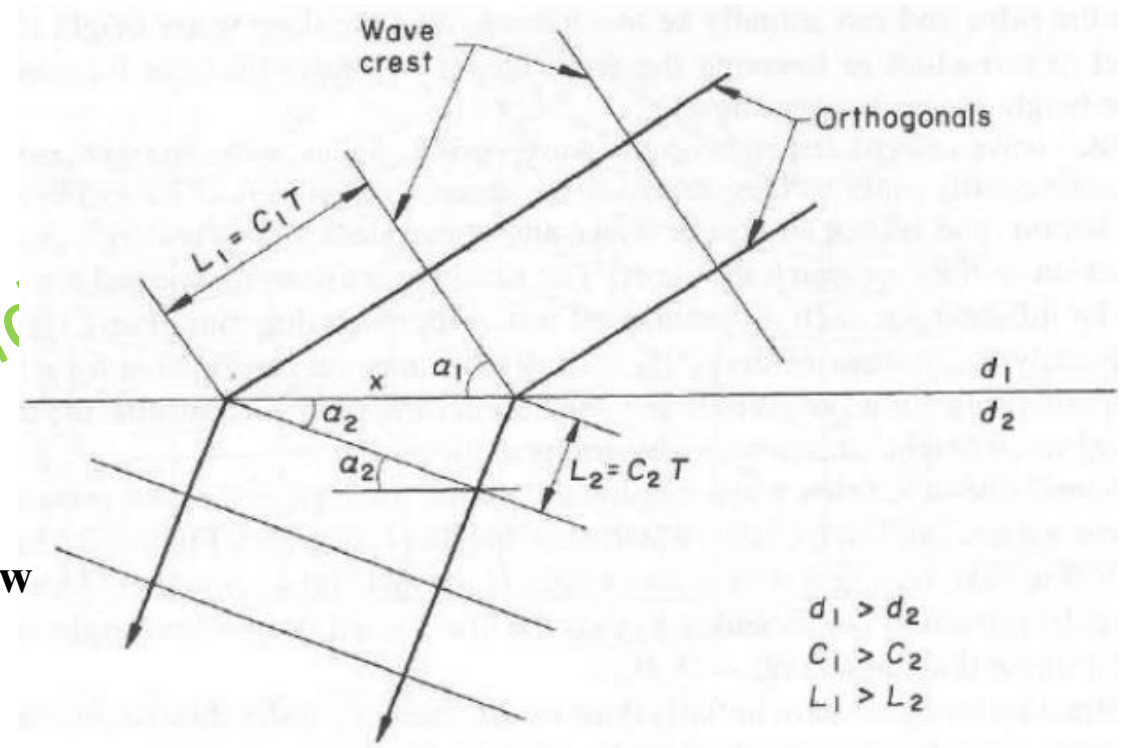


Figure-7: Definition sketch for Snell's law

- Waves approaching at an **oblique angle** are refracted in such a way that the **angle α** is decreased, corresponding to the **wave fronts** that tend to become more nearly **parallel to the shore**.

- Let us consider two wave rays, defined as **orthogonal** or **ray of the wavefronts**, at a distance b apart at constant

$$\frac{d}{L_0} = 0.5,$$

$$\cos \alpha_0 = \frac{b_0}{BC}$$

$$\cos \alpha = \frac{b}{BC}$$

$$BC = \frac{b_0}{\cos \alpha_0} = \frac{b}{\cos \alpha}$$

$$\boxed{\frac{\cos \alpha_0}{\cos \alpha} = \frac{b_0}{b}}$$

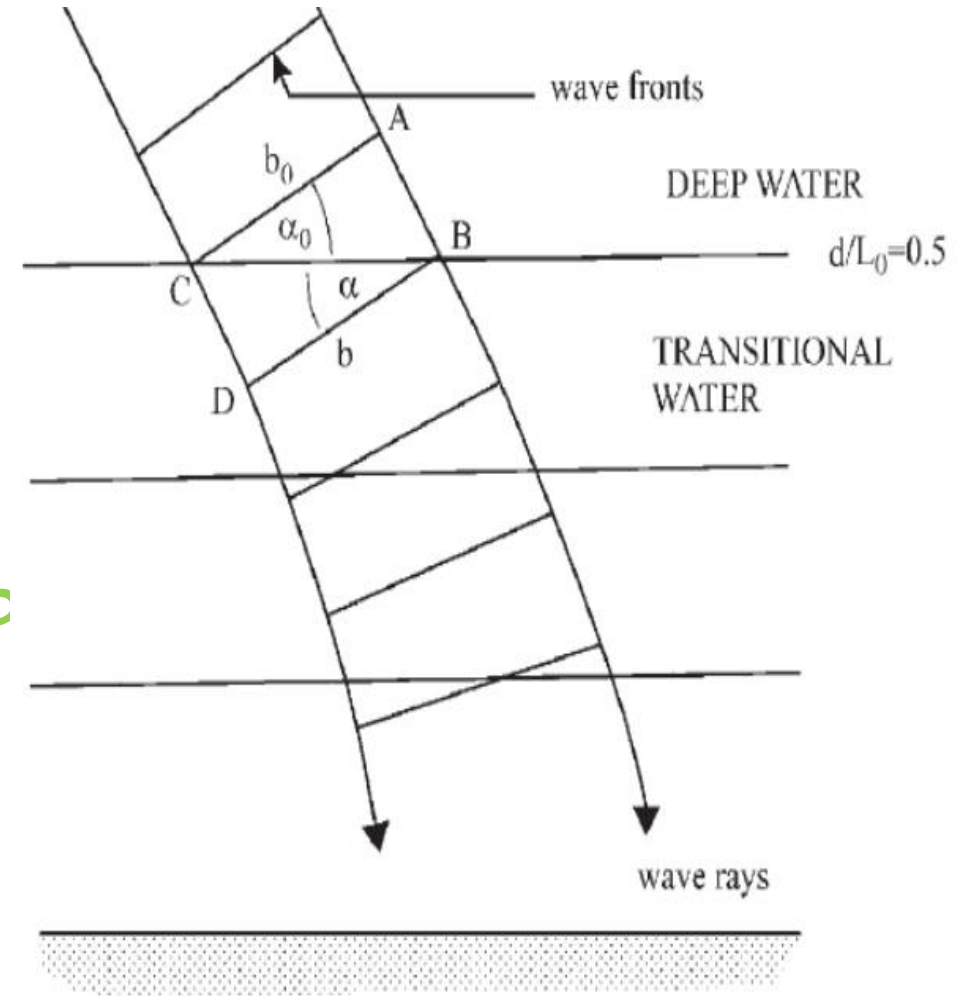


Fig 8. Schematic representation of wave deformation

- A **refraction diagram** is a plot showing the coordinate positions of the **orthogonal** for a given location with defined **bathymetry** (depth contours).
- The figure shows a typical view of the understanding of **wave refraction**.

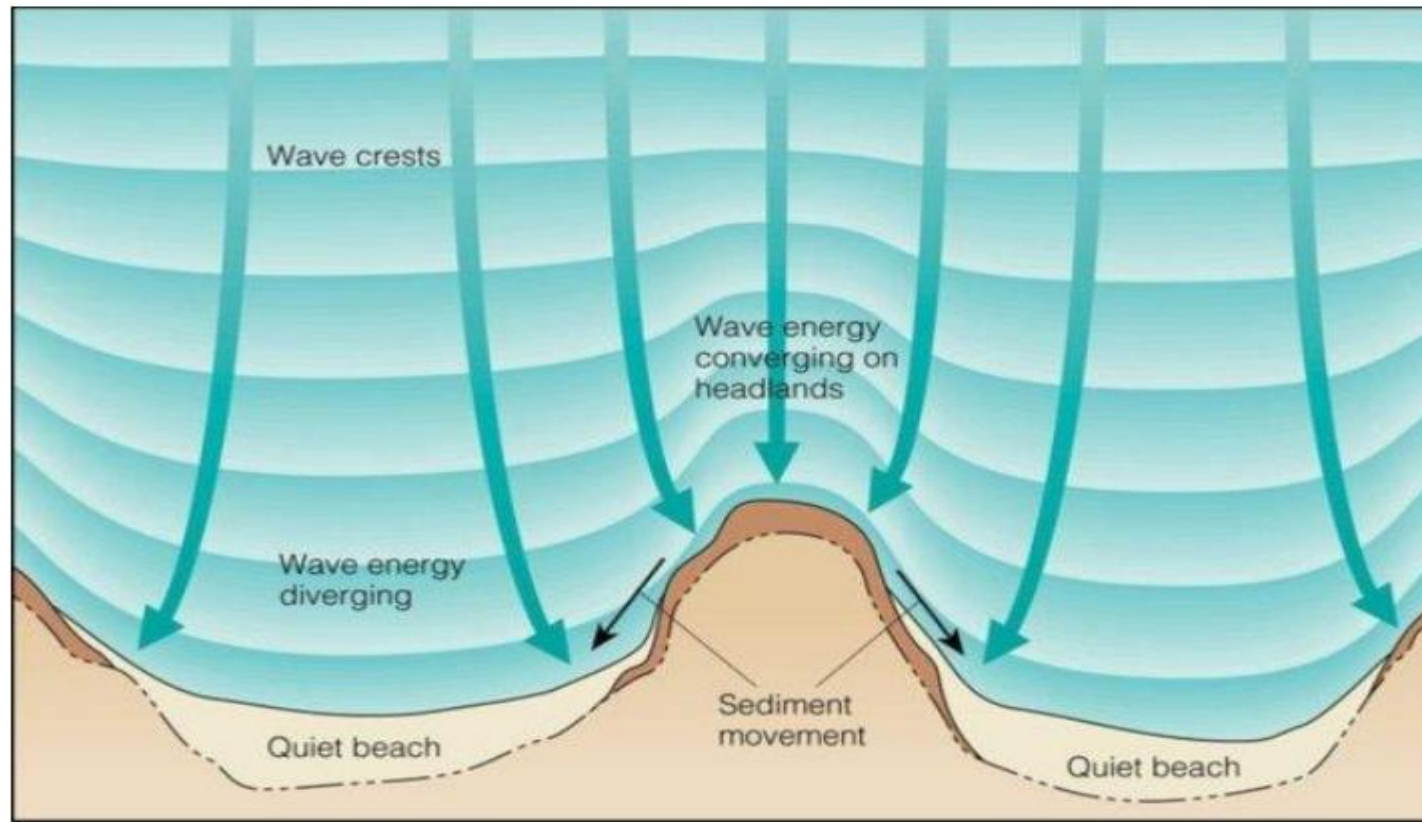


Fig 9. Wave refraction diagram

- **Wave energy converges** in the case of **convex depth contours** and **diverges** in the case of **concave depth contours**.
- **Convergence** of orthogonal leads to an increase in **wave heights** or **energy**, resulting in **shore erosion**.
- **Divergence** of orthogonal leads to a reduction in **wave heights** or **energy**, resulting in **accretion or deposition**.

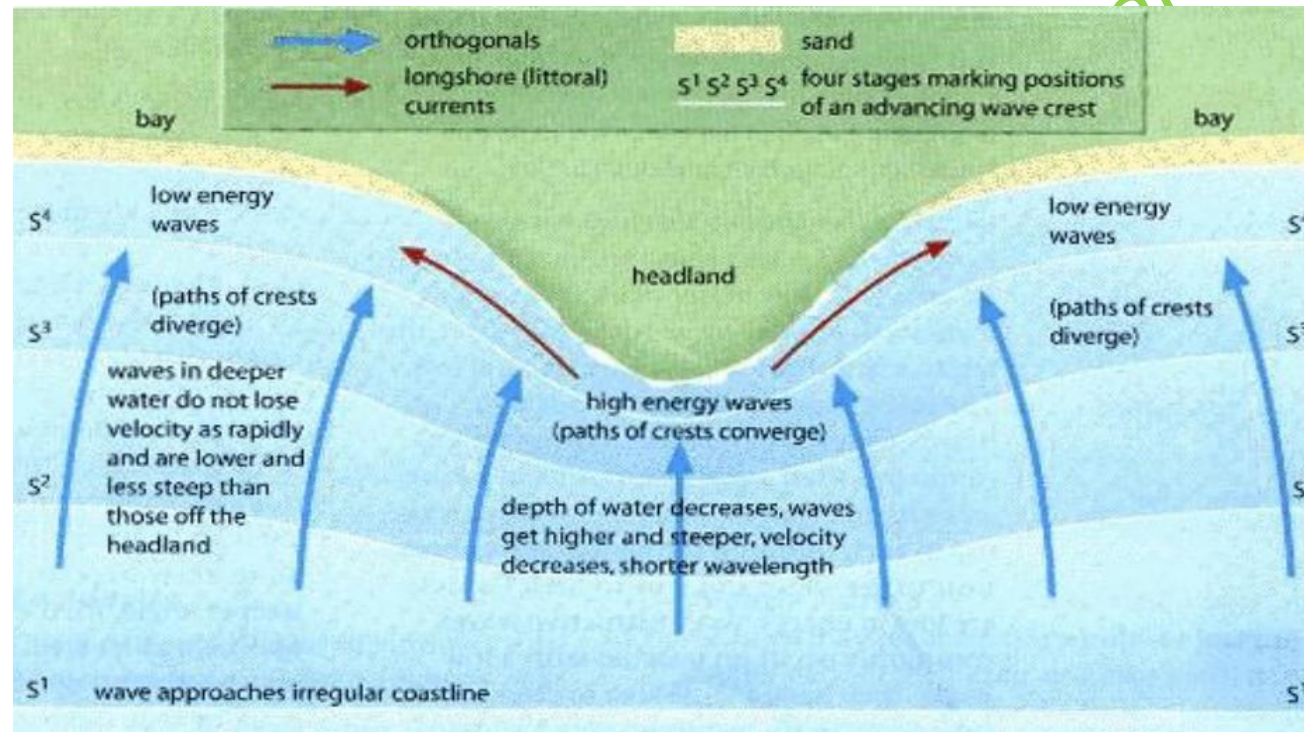


Fig 10. Wave refraction diagram

Combined effect of refraction and shoaling

- In the analysis of the **refraction phenomenon**, it is assumed that for an advancing wave approaching the shore, energy flow is **absent** along the wave crest in the lateral direction.
- It implies that the energy transmitted remains constant between the orthogonal.
- The **average power** transmitted by a wave is given by

$$P = nbEC$$

$$\text{where, } n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right]$$

Deep water conditions

$$\frac{2kd}{\sinh 2kd} = 0 \text{ and } C_g = \frac{1}{2}c \text{ and } n = \frac{1}{2}$$

$$P_0 = \frac{1}{2} b_0 E_0 C_0$$

Shallow water conditions

$$\sinh 2kd = 2kd \text{ and } C_g = nC$$

$$P = nbEC$$

- Since energy flow is absent along the wave crest in the lateral direction

$$P = P_0$$

$$nbEC = \frac{1}{2} b_0 E_0 C_0$$

$$\frac{E}{E_0} = \frac{1}{2n} \frac{b_0}{b} \frac{C_0}{C} \quad (i)$$

$$\frac{E}{E_0} = \frac{\frac{1}{8} \rho g H^2}{\frac{1}{8} \rho g H_0^2} \quad (ii)$$

From Equation (i) and (ii), we found as:

$$\frac{\frac{1}{8} \rho g H^2}{\frac{1}{8} \rho g H_0^2} = \frac{1}{2n} \frac{b_0}{b} \frac{C_0}{C}$$

$$\boxed{\frac{H}{H_0} = \sqrt{\frac{1}{2n} \frac{C_0}{C}} \sqrt{\frac{b_0}{b}} = K_s K_r}$$

Shoaling Co-efficient (K_s)

Refraction Co-efficient (K_r)

Problem

A deep water wave of height 3.5 m and period 10 s is refracted so that the distance between the orthogonals is reduced by 50 % at a depth of 10 m and reduced by 20 % at 5 m water depth. What will be the height of the wave here, assuming no energy loss?

Solution- Given, $H_0 = 3.5\text{m}$, $T = 10\text{s}$

H in depth of 10m

$$\frac{H}{H_0} = K_s \cdot K_r = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{C}} \cdot \sqrt{\frac{b_0}{b}}$$

$$\sqrt{\frac{b_0}{b}} = \sqrt{\frac{100}{50}} = \sqrt{2}$$

(as orthogonal is reduced by 50%)

$$\frac{H}{H_0} = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{C}} \cdot \sqrt{\frac{b_0}{b}} = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{C}} \cdot \sqrt{2}$$

$$L_0 = 1.56T^2 = 1.56 \times 10^2 = 156\text{m}$$

$$\frac{d}{L_0} = \frac{10}{156} = 0.0642$$

$$\frac{d}{L} = 0.1082 \text{ (from wave table)}$$

$$\frac{10}{L} = 0.1082 \quad \text{or, } L = 92.42 \text{ m}$$

$$C_0 = \frac{L_0}{T} = \frac{156}{10} = 15.6 \text{ m/s}$$

$$C = \frac{L}{T} = \frac{92.4}{10} = 9.24 \text{ m/s}$$

$$n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right] = 0.88$$

$$k = \frac{2\pi}{L} = 0.06523$$

d/L ₀	d/L	2πd/L	tanh (2πd/L)	sinh (2πd/L)	cosh (2πd/L)	H/H' ₀	K	4πd/L	sinh (4πd/L)	cosh (4πd/L)	n	C _G /C ₀	M
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0.0700	0.1139	0.7159	0.6144	0.7786	1.2674	0.9713	0.7890	1.4318	1.9737	2.2126	0.8627	0.5300	13.07

$$K_s = \sqrt{\frac{C_0}{c} \cdot \frac{1}{2n}} = 0.98$$

$$\frac{H}{H_0} = 0.98 \times \sqrt{2}$$

$$H = 0.98 \times \sqrt{2} \times 3.5$$

$$H = 4.85\text{m}$$

H in depth of 5 m

$$\frac{H}{H_0} = K_s \cdot K_r$$

$$\frac{H}{H_0} = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{c}} \cdot \sqrt{\frac{b_0}{b}} = \sqrt{\frac{1}{2n} \cdot \frac{C_0}{c}} \cdot \sqrt{\frac{10}{8}}$$

$$L_0 = 1.56 T^2 = 1.56 \times 10^2 = 156\text{m}$$

$$\frac{d}{L_0} = \frac{5}{156} = 0.032$$

$$\frac{d}{L} = 0.0738 \text{ (from wave table)}$$

$$\frac{5}{L} = 0.0738 \quad \text{or, } L = 67.75 \text{ m}$$

$$C_0 = \frac{L_0}{T} = \frac{156}{10} = 15.6 \text{ m/s}$$

$$C = \frac{L}{T} = \frac{67.75}{10} = 6.78 \text{ m/s}$$

$$n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right] = 0.935$$

$$K_s = \sqrt{\frac{C_0}{c} \cdot \frac{1}{2n}} = 1.109$$

d/L ₀	d/L	2πd/L	tanh (2πd/L)	sinh (2πd/L)	cosh (2πd/L)	H/H' ₀	K	4πd/L	sinh (4πd/L)	cosh (4πd/L)	n	C ₀ /C ₀	M
0.0310	0.0726	0.4562	0.4270	0.4722	1.1059	1.1180	0.9043	0.9124	1.0443	1.4459	0.9368	0.4000	27.07
0.0320	0.0738	0.4640	0.4333	0.4808	1.1096	1.1110	0.9012	0.9280	1.0670	1.4624	0.9348	0.4051	26.28
0.0330	0.0751	0.4717	0.4396	0.4894	1.1133	1.1042	0.8982	0.9434	1.0897	1.4790	0.9329	0.4101	25.54
0.0340	0.0763	0.4793	0.4457	0.4979	1.1171	1.0978	0.8952	0.9586	1.1124	1.4958	0.9309	0.4149	24.84
0.0350	0.0775	0.4868	0.4517	0.5063	1.1209	1.0916	0.8922	0.9737	1.1350	1.5127	0.9289	0.4196	24.19
0.0360	0.0787	0.4943	0.4576	0.5147	1.1247	1.0857	0.8892	0.9886	1.1577	1.5298	0.9270	0.4242	23.57
0.0370	0.0798	0.5017	0.4634	0.5230	1.1285	1.0800	0.8861	1.0033	1.1803	1.5470	0.9250	0.4287	22.98
0.0380	0.0810	0.5089	0.4691	0.5312	1.1323	1.0745	0.8831	1.0179	1.2030	1.5644	0.9231	0.4330	22.42
0.0390	0.0822	0.5162	0.4747	0.5394	1.1362	1.0693	0.8801	1.0323	1.2257	1.5819	0.9211	0.4373	21.90
0.0400	0.0833	0.5233	0.4803	0.5475	1.1401	1.0643	0.8771	1.0466	1.2485	1.5996	0.9192	0.4414	21.40
0.0410	0.0844	0.5304	0.4857	0.5556	1.1440	1.0594	0.8741	1.0608	1.2713	1.6174	0.9172	0.4455	20.92
0.0420	0.0855	0.5374	0.4910	0.5637	1.1479	1.0548	0.8711	1.0749	1.2941	1.6355	0.9153	0.4494	20.47
0.0430	0.0866	0.5444	0.4963	0.5717	1.1519	1.0503	0.8682	1.0888	1.3170	1.6536	0.9134	0.4533	20.03
0.0440	0.0877	0.5513	0.5015	0.5796	1.1559	1.0459	0.8652	1.1026	1.3399	1.6720	0.9114	0.4571	19.62
0.0450	0.0888	0.5581	0.5066	0.5876	1.1598	1.0417	0.8622	1.1163	1.3630	1.6905	0.9095	0.4607	19.23
0.0460	0.0899	0.5649	0.5116	0.5955	1.1639	1.0377	0.8592	1.1299	1.3861	1.7091	0.9076	0.4643	18.85
0.0470	0.0910	0.5717	0.5166	0.6033	1.1679	1.0338	0.8562	1.1433	1.4092	1.7280	0.9057	0.4678	18.49
0.0480	0.0920	0.5784	0.5215	0.6111	1.1720	1.0300	0.8533	1.1567	1.4325	1.7470	0.9037	0.4713	18.15
0.0490	0.0931	0.5850	0.5263	0.6189	1.1761	1.0264	0.8503	1.1700	1.4558	1.7662	0.9018	0.4746	17.82
0.0500	0.0942	0.5916	0.5310	0.6267	1.1802	1.0229	0.8473	1.1832	1.4792	1.7855	0.8999	0.4779	17.50

$$\frac{H}{H_0} = K_s \cdot K_r = 1.109 \times 1.118$$

$$H = 1.109 \times 1.118 \times 3.5 = 4.34 \text{ m}$$

- In this case, as per the given problem, H in d = 5 m is 4.34, which is greater than the breaking wave height in this depth of $0.78 \times 5 = 3.9$ m. (as maximum wave height 0.78d)
- This only suggests that the 20% reduction in 5 m water depth would not have wave of height greater than 3.9 m.

Dr. Mohammad Saud Afzal

Problem

For an average deep-water wave height of 1.5 m, $T_{av} = 10$ s and $\theta_{o, av} = 50^\circ$, determine the length of the wave energy device if it is to be located in water depth of 6 m and power required is 200 kW. Assume that the device is aligned parallel to the wave crest.

Solution:

Given: $H_o = 1.5$ m, $d = 6$ m, $\theta_o = 50^\circ$, $T = 10$ s

$$L_o = 1.56T^2 = 1.56 \times 10^2 = 156 \text{ m}$$

$$\frac{d}{L_o} = \frac{6}{156} = 0.0384$$

$$\frac{d}{L} = 0.0810 \text{ (from wave table)}$$

$$\frac{6}{L} = 0.0810 \quad \text{or, } L = 74.07 \text{ m}$$

$$C_o = L_o/T = 156/10 = 15.6 \text{ m/s}$$

$$C = L/T = 74.07/10 = 7.41 \text{ m/s}$$

d/L_o	d/L	$2\pi d/L$	$\tanh(2\pi d/L)$	$\sinh(2\pi d/L)$	$\cosh(2\pi d/L)$	H/H_o	K	$4\pi d/L$	$\sinh(4\pi d/L)$	$\cosh(4\pi d/L)$	n	C_g/C_o	M
0.0310	0.0726	0.4562	0.4270	0.4722	1.1059	1.1180	0.9043	0.9124	1.0443	1.4459	0.9368	0.4000	27.07
0.0320	0.0738	0.4640	0.4333	0.4808	1.1096	1.1110	0.9012	0.9280	1.0670	1.4624	0.9348	0.4051	26.28
0.0330	0.0751	0.4717	0.4396	0.4894	1.1133	1.1042	0.8982	0.9434	1.0897	1.4790	0.9329	0.4101	25.54
0.0340	0.0763	0.4793	0.4457	0.4979	1.1171	1.0978	0.8952	0.9586	1.1124	1.4958	0.9309	0.4149	24.84
0.0350	0.0775	0.4868	0.4517	0.5063	1.1209	1.0916	0.8922	0.9737	1.1350	1.5127	0.9289	0.4196	24.19
0.0360	0.0787	0.4943	0.4576	0.5147	1.1247	1.0857	0.8892	0.9886	1.1577	1.5298	0.9270	0.4242	23.57
0.0370	0.0798	0.5017	0.4634	0.5230	1.1285	1.0800	0.8861	1.0033	1.1803	1.5470	0.9250	0.4287	22.98
0.0380	0.0810	0.5089	0.4691	0.5312	1.1323	1.0745	0.8831	1.0179	1.2030	1.5644	0.9231	0.4330	22.42
0.0390	0.0822	0.5162	0.4747	0.5394	1.1362	1.0693	0.8801	1.0323	1.2257	1.5819	0.9211	0.4373	21.90
0.0400	0.0833	0.5233	0.4803	0.5475	1.1401	1.0643	0.8771	1.0466	1.2485	1.5996	0.9192	0.4414	21.40
0.0410	0.0844	0.5304	0.4857	0.5556	1.1440	1.0594	0.8741	1.0608	1.2713	1.6174	0.9172	0.4455	20.92
0.0420	0.0855	0.5374	0.4910	0.5637	1.1479	1.0548	0.8711	1.0749	1.2941	1.6355	0.9153	0.4494	20.47
0.0430	0.0866	0.5444	0.4963	0.5717	1.1519	1.0503	0.8682	1.0888	1.3170	1.6536	0.9134	0.4533	20.03
0.0440	0.0877	0.5513	0.5015	0.5796	1.1559	1.0459	0.8652	1.1026	1.3399	1.6720	0.9114	0.4571	19.62
0.0450	0.0888	0.5581	0.5066	0.5876	1.1598	1.0417	0.8622	1.1163	1.3630	1.6905	0.9095	0.4607	19.23
0.0460	0.0899	0.5649	0.5116	0.5955	1.1639	1.0377	0.8592	1.1299	1.3861	1.7091	0.9076	0.4643	18.85
0.0470	0.0910	0.5717	0.5166	0.6033	1.1679	1.0338	0.8562	1.1433	1.4092	1.7280	0.9057	0.4678	18.49
0.0480	0.0920	0.5784	0.5215	0.6111	1.1720	1.0300	0.8533	1.1567	1.4325	1.7470	0.9037	0.4713	18.15
0.0490	0.0931	0.5850	0.5263	0.6189	1.1761	1.0264	0.8503	1.1700	1.4558	1.7662	0.9018	0.4746	17.82
0.0500	0.0942	0.5916	0.5310	0.6267	1.1802	1.0229	0.8473	1.1832	1.4792	1.7855	0.8999	0.4779	17.50

From Snell's law, $\frac{\sin\theta_o}{C_o} = \frac{\sin\theta}{C}$

$$\frac{\sin 50^\circ}{15.6} = \frac{\sin\theta}{7.36} \text{ leading to } \theta = 21.18^\circ$$

$$\frac{H}{H_o} = \sqrt{\frac{C_o}{2nC}} \cdot \sqrt{\frac{\cos\theta_o}{\cos\theta}}$$

$$\frac{H}{H_o} = \sqrt{\frac{15.6}{2 \times 0.922 \times 7.36}} \cdot \sqrt{\frac{\cos 50^\circ}{\cos 21.18^\circ}} = 0.889$$

$$H = 0.889 \times 1.5 = 1.335 \text{ m.}$$

$$P = (\gamma H^2 W / 8) \cdot C_g$$

(where W is the width of the wave energy device)

$$200,000 = (1030 \times 9.81 \times 1.335^2 \times W / 8) \times (0.922 \times 7.36)$$

Width of the wave energy device should be $(W) = 13.09 \text{ m.}$

Diffraction

- **Diffraction** is the bending or spreading of waves when they encounter an obstacle or pass through an opening.
- It is dominant around **natural barriers or man-made structures** such as **breakwaters, groins, training walls, etc.**

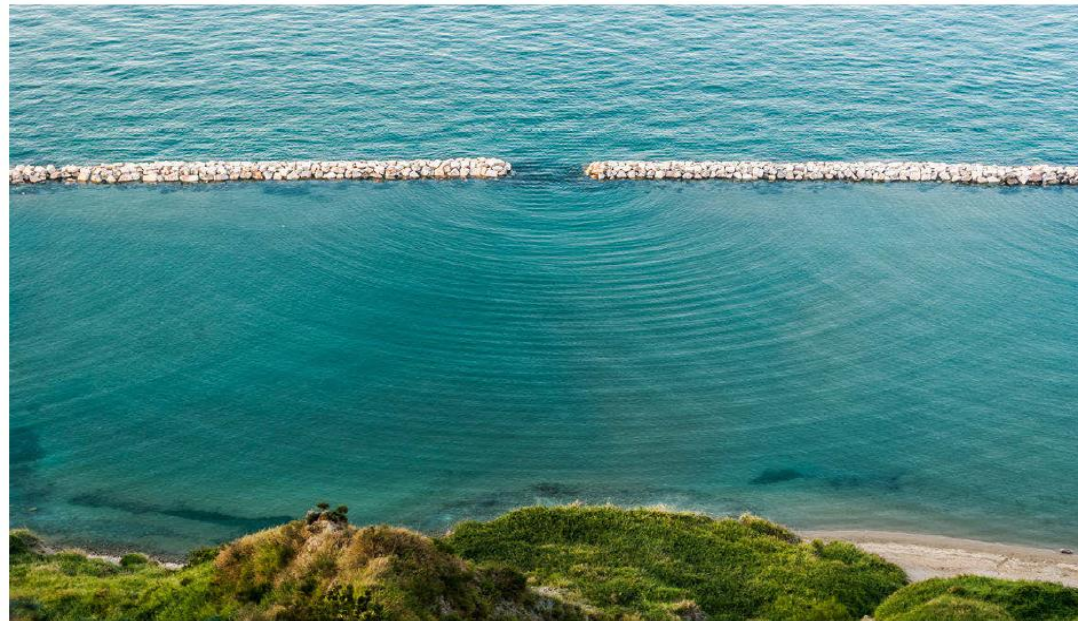
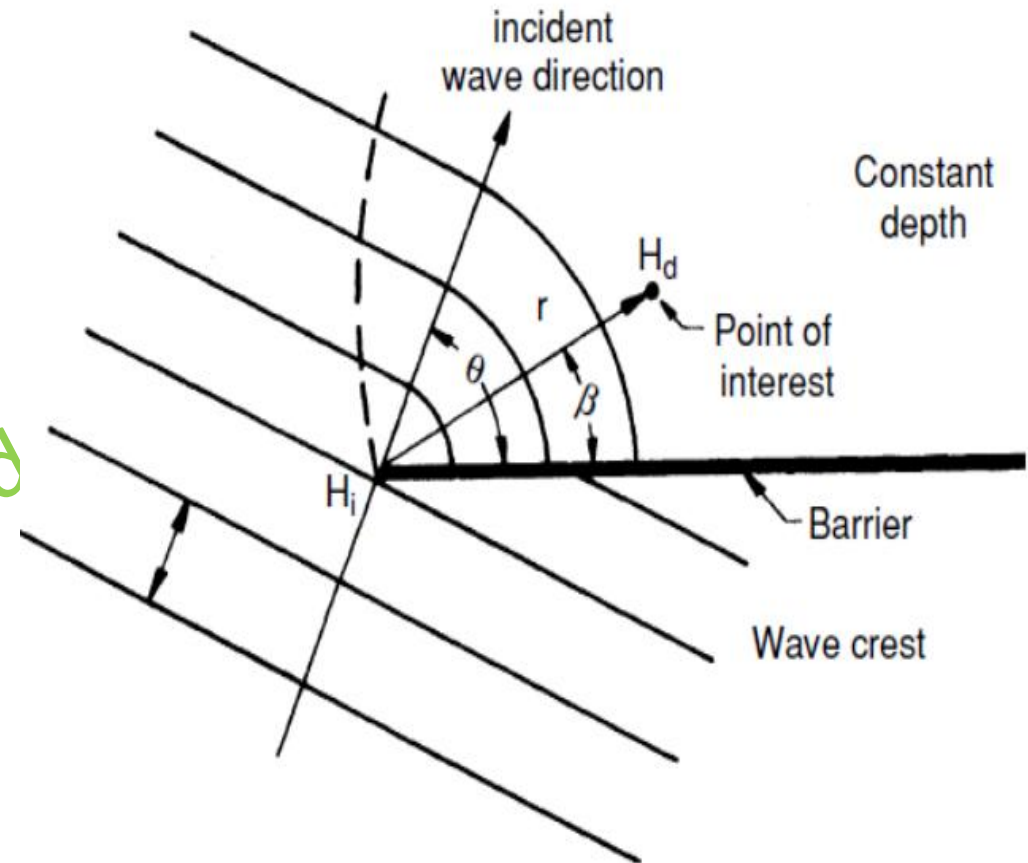


Fig 11. Typical diffraction pattern through a gap between two detached breakwaters in Pesaro, Italy

Wave diffraction coefficient:

- If H_i is the incident wave height at the end of the barrier and H_d is the height of a wave diffracted at the point of interest in a sheltered area, then the diffraction coefficient (K_d) could be defined as:

$$K_d = \frac{H_d}{H_i}$$



Calculation of wave diffraction coefficient:

- The value of K_d depends on the location behind the barrier defined by r and β , and the incident wave direction defined by θ ; or in the dimensionless form given by

$$K_d = f\left(\frac{r}{L}, \beta, \theta\right)$$

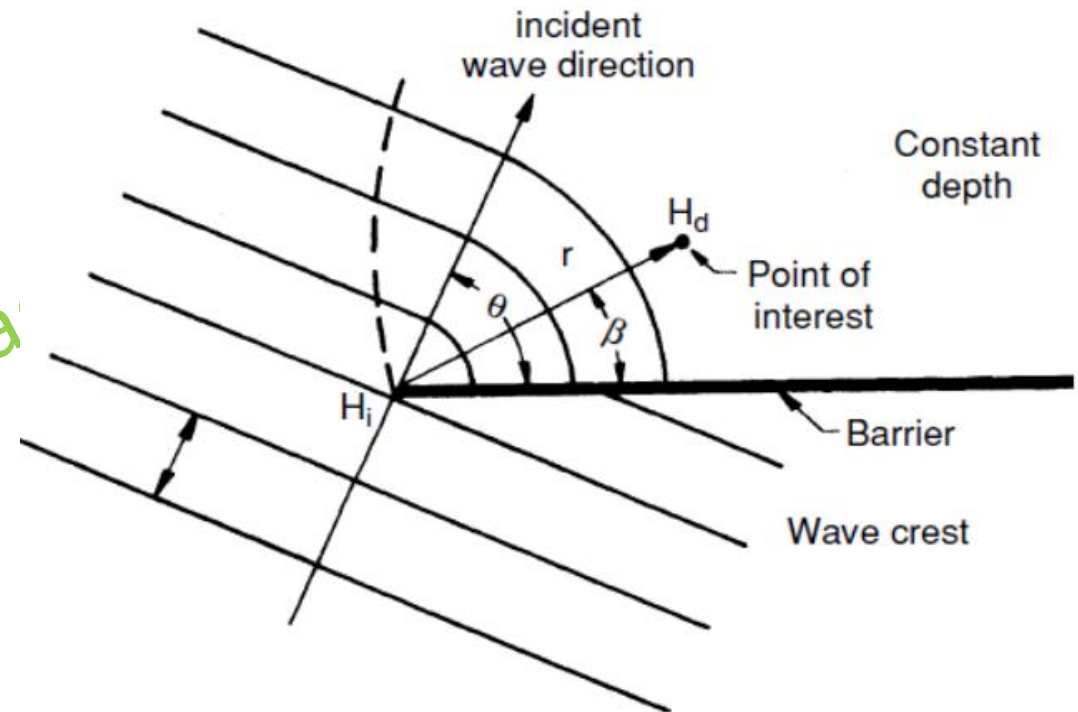
Where

r = Distance of point of interest

β = Diffracted wave direction from barrier

θ = Incident wave direction from barrier

L = Wavelength in the lee of the barrier



The two most common diffraction problems encountered in coastal engineering design are:

- Diffraction past the end of a **semi-infinite barrier**, and
- Diffraction through a relatively small **gap in a barrier**.

Semi-infinite Barrier:

- The diffraction solution for a semi-infinite barrier is presented by Wiegel (1964).
- Penny and Price (1952) showed that the mathematical solution for the diffraction of light can be used to predict the wave crest pattern and height variation for these two wave diffraction problems.
- Wiegel (1962) used the Penny and Price (1952) solution to calculate and tabulate values of K_d for selected values of θ , β , and r/L .
- The values of K_d against these parameters are shown in the next slide, as reported in Wiegel (1962).

K_d versus $\frac{r}{L}$, β , θ for Semi-Infinite Breakwater

r/L	β (Degrees)												
	0	15	30	45	60	75	90	105	120	135	150	165	180
$\theta = 15^\circ$													
1/2	0.49	0.79	0.83	0.90	0.97	1.01	1.03	1.02	1.01	0.99	0.99	1.00	1.00
1	0.38	0.73	0.83	0.95	1.04	1.04	0.99	0.98	1.01	1.01	1.00	1.00	1.00
2	0.21	0.68	0.86	1.05	1.03	0.97	1.02	0.99	1.00	1.00	1.00	1.00	1.00
5	0.13	0.63	0.99	1.04	1.03	1.02	0.99	0.99	1.00	1.01	1.00	1.00	1.00
10	0.35	0.58	1.10	1.05	0.98	0.99	1.01	1.00	1.00	1.00	1.00	1.00	1.00
$\theta = 30^\circ$													
1/2	0.61	0.63	0.68	0.76	0.87	0.97	1.03	1.05	1.03	1.01	0.99	0.95	1.00
1	0.50	0.53	0.63	0.78	0.95	1.06	1.05	0.98	0.98	1.01	1.01	0.97	1.00
2	0.40	0.44	0.59	0.84	1.07	1.03	0.96	1.02	0.98	1.01	0.99	0.95	1.00
5	0.27	0.32	0.55	1.00	1.04	1.04	1.02	0.99	0.99	1.00	1.01	0.97	1.00
10	0.20	0.24	0.54	1.12	1.06	0.97	0.99	1.01	1.00	1.00	1.00	0.98	1.00
$\theta = 45^\circ$													
1/2	0.49	0.50	0.55	0.63	0.73	0.85	0.96	1.04	1.06	1.04	1.00	0.99	1.00
1	0.38	0.40	0.47	0.59	0.76	0.95	1.07	1.06	0.98	0.97	1.01	1.01	1.00
2	0.29	0.31	0.39	0.56	0.83	1.08	1.04	0.96	1.03	0.98	1.01	1.00	1.00
5	0.18	0.20	0.29	0.54	1.01	1.04	1.05	1.03	1.00	0.99	1.01	1.00	1.00
10	0.13	0.15	0.22	0.53	1.13	1.07	0.96	0.98	1.02	0.99	1.00	1.00	1.00

K_d versus $\frac{r}{L}$, β , θ for Semi-Infinite Breakwater

r/L	β (Degrees)												
	0	15	30	45	60	75	90	105	120	135	150	165	180
$\theta = 60^\circ$													
1/2	0.40	0.41	0.45	0.52	0.60	0.72	0.85	1.13	1.04	1.06	1.03	1.01	1.00
1	0.31	0.32	0.36	0.44	0.57	0.75	0.96	1.08	1.06	0.98	0.98	1.01	1.00
2	0.22	0.23	0.28	0.37	0.55	0.83	1.08	1.04	0.96	1.03	0.98	1.01	1.00
5	0.14	0.15	0.18	0.28	0.53	1.01	1.04	1.05	1.03	0.99	0.99	1.00	1.00
10	0.10	0.11	0.13	0.21	0.52	1.14	1.07	0.96	0.98	1.01	1.00	1.00	1.00
$\theta = 75^\circ$													
1/2	0.34	0.35	0.38	0.42	0.50	0.59	0.71	0.85	0.97	1.04	1.05	1.02	1.00
1	0.25	0.26	0.29	0.34	0.43	0.56	0.75	0.95	1.02	1.06	0.98	0.98	1.00
2	0.18	0.19	0.22	0.26	0.36	0.54	0.83	1.09	1.04	0.96	1.03	0.99	1.00
5	0.12	0.12	0.13	0.17	0.27	0.52	1.01	1.04	1.05	1.03	0.99	0.99	1.00
10	0.08	0.08	0.10	0.13	0.20	0.52	1.14	1.07	0.96	0.98	1.01	1.00	1.00
$\theta = 90^\circ$													
1/2	0.31	0.31	0.33	0.36	0.41	0.49	0.59	0.71	0.85	0.96	1.03	1.03	1.00
1	0.22	0.23	0.24	0.28	0.33	0.42	0.56	0.75	0.96	1.07	1.05	0.99	1.00
2	0.16	0.16	0.18	0.20	0.26	0.35	0.54	0.69	1.08	1.04	0.96	1.02	1.00
5	0.10	0.10	0.11	0.13	0.16	0.27	0.53	1.01	1.04	1.05	1.02	0.99	1.00
10	0.07	0.07	0.08	0.09	0.13	0.20	0.52	1.14	1.07	0.96	0.99	1.01	1.00

K_d versus $\frac{r}{L}$, β , θ for Semi-Infinite Breakwater

$\theta = 105^\circ$													
1/2	0.28	0.28	0.29	0.32	0.35	0.41	0.49	0.59	0.72	0.85	0.97	1.01	1.00
1	0.20	0.20	0.24	0.23	0.27	0.33	0.42	0.56	0.75	0.95	1.06	1.04	1.00
2	0.14	0.14	0.13	0.17	0.20	0.25	0.35	0.54	0.83	1.08	1.03	0.97	1.00
5	0.09	0.09	0.10	0.11	0.13	0.17	0.27	0.52	1.02	1.04	1.04	1.02	1.00
10	0.07	0.06	0.08	0.08	0.09	0.12	0.20	0.52	1.14	1.07	0.97	0.99	1.00
$\theta = 120^\circ$													
1/2	0.25	0.26	0.27	0.28	0.31	0.35	0.41	0.50	0.60	0.73	0.87	0.97	1.00
1	0.18	0.19	0.19	0.21	0.23	0.27	0.33	0.43	0.57	0.76	0.95	1.04	1.00
2	0.13	0.13	0.14	0.14	0.17	0.20	0.26	0.16	0.55	0.83	1.07	1.03	1.00
5	0.08	0.08	0.08	0.09	0.11	0.13	0.16	0.27	0.53	1.01	1.04	1.03	1.00
10	0.06	0.06	0.06	0.07	0.07	0.09	0.13	0.20	0.52	1.13	1.06	0.98	1.00
$\theta = 135^\circ$													
1/2	0.24	0.24	0.25	0.26	0.28	0.32	0.36	0.42	0.52	0.63	0.76	0.90	1.00
1	0.18	0.17	0.18	0.19	0.21	0.23	0.28	0.34	0.44	0.59	0.78	0.95	1.00
2	0.12	0.12	0.13	0.14	0.14	0.17	0.20	0.26	0.37	0.56	0.84	1.05	1.00
5	0.08	0.07	0.08	0.08	0.09	0.11	0.13	0.17	0.28	0.54	1.00	1.04	1.00
10	0.05	0.06	0.06	0.06	0.07	0.08	0.09	0.13	0.21	0.53	1.12	1.05	1.00
$\theta = 150^\circ$													
1/2	0.23	0.23	0.24	0.25	0.27	0.29	0.33	0.38	0.45	0.55	0.68	0.83	1.00
1	0.16	0.17	0.17	0.18	0.19	0.22	0.24	0.29	0.36	0.47	0.63	0.83	1.00

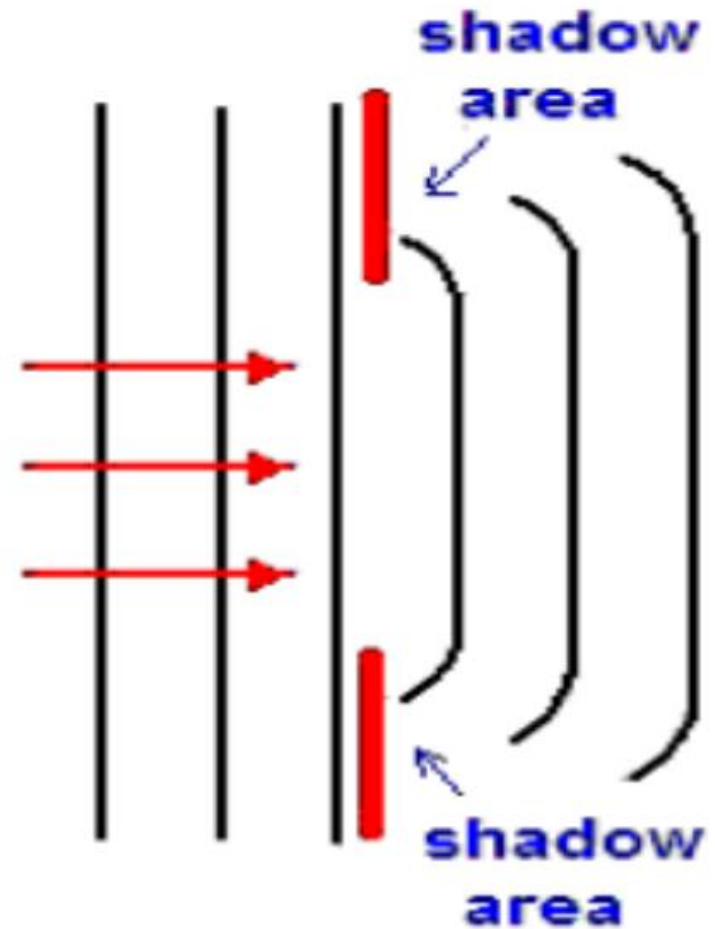
K_d versus $\frac{r}{L}$, β , θ for Semi-Infinite Breakwater

r/L	β (Degrees)												
	0	15	30	45	60	75	90	105	120	135	150	165	180
2	0.12	0.12	0.12	0.13	0.14	0.15	0.18	0.22	0.28	0.39	0.59	0.86	1.00
5	0.07	0.07	0.08	0.08	0.08	0.10	0.11	0.13	0.18	0.29	0.55	0.99	1.00
10	0.05	0.05	0.05	0.06	0.06	0.07	0.08	0.10	0.13	0.22	0.54	1.10	1.00
$\theta = 165^\circ$													
1/2	0.23	0.23	0.23	0.24	0.26	0.28	0.31	0.35	0.41	0.50	0.63	0.79	1.00
1	0.16	0.16	0.17	0.17	0.19	0.20	0.23	0.26	0.32	0.40	0.53	0.73	1.00
2	0.11	0.11	0.12	0.12	0.13	0.14	0.16	0.19	0.23	0.31	0.44	0.68	1.00
5	0.07	0.07	0.07	0.07	0.08	0.09	0.10	0.12	0.15	0.20	0.32	0.63	1.00
10	0.05	0.05	0.05	0.06	0.06	0.06	0.07	0.08	0.11	0.11	0.21	0.58	1.00
$\theta = 180^\circ$													
1/2	0.20	0.25	0.23	0.24	0.25	0.28	0.31	0.34	0.40	0.49	0.61	0.78	1.00
1	0.10	0.17	0.16	0.18	0.18	0.23	0.22	0.25	0.31	0.38	0.50	0.70	1.00
2	0.02	0.09	0.12	0.12	0.13	0.18	0.16	0.18	0.22	0.29	0.40	0.60	1.00
5	0.02	0.06	0.07	0.07	0.07	0.08	0.10	0.12	0.14	0.18	0.27	0.46	1.00
10	0.01	0.05	0.05	0.06	0.06	0.07	0.07	0.08	0.10	0.13	0.20	0.36	1.00

From Wiegel, 1962.

Barrier Gap: Large

- If the **gap is large**, compared to the wavelength of the waves passing through, the **circular disturbances** are tiny compared to the undisturbed wavefronts.
- A **tiny disturbance** at the **edge** of the straight wavefront leads to slight curving of the wavefronts at the edge and a slight spread.



Barrier Gap: Small

- If the **gap is small**, the circular disturbances that get through are massive compared to the undisturbed wavefronts.
- **only circular wavefronts** are observed passing through the tiny gap in the barrier (instead of straight ones)



Energy that gets through spreads out behind the barrier too.

Problem

Consider a train of 6 s period waves approaching a breakwater so that the angle of approach at the breakwater head is 60° . The water depth in the lee of the breakwater is 10 m. Determine the wave height at an angle of 30° from the breakwater and a distance of 96.6 m from the breakwater head if the incident wave height at the head is 2.2 m.

Solution- Given $T=6$ sec , $\theta = 60^\circ$, $\beta = 30^\circ$, $r = 96.6$, $d = 10$, $H_i=2.2$, $H_d = ?$

$$L_0 = 1.56 \times T^2 = 1.56 \times 6^2 = 56.16 \text{ m}$$

$$\frac{d}{L_0} = \frac{10}{56.16} = 0.178$$

$$\frac{d}{L} = 0.2067 \text{ (from wave table)}$$

$$\frac{10}{L} = 0.2067$$

$$L = 48.3 \text{ m}$$

$$\frac{r}{L} = \frac{96.6}{48.3} = 2$$

d/L_0	d/L	$2\pi d/L$	$\tanh(2\pi d/L)$	$\sinh(2\pi d/L)$	$\cosh(2\pi d/L)$	H/H_0	K	$4\pi d/L$	$\sinh(4\pi d/L)$	$\cosh(4\pi d/L)$	n	C_g/C_0	M
0.1710	0.2008	1.2617	0.8515	1.6242	1.9074	0.9135	0.5243	2.5235	6.1959	6.2760	0.7036	0.5992	6.81
0.1720	0.2016	1.2670	0.8530	1.6342	1.9159	0.9136	0.5219	2.5340	6.2620	6.3414	0.7023	0.5991	6.78
0.1730	0.2025	1.2722	0.8544	1.6443	1.9245	0.9137	0.5196	2.5445	6.3289	6.4074	0.7010	0.5990	6.76
0.1740	0.2033	1.2775	0.8558	1.6544	1.9332	0.9138	0.5173	2.5550	6.3965	6.4742	0.6997	0.5988	6.74
0.1750	0.2042	1.2827	0.8572	1.6646	1.9419	0.9139	0.5150	2.5654	6.4647	6.5416	0.6984	0.5987	6.72
0.1760	0.2050	1.2880	0.8586	1.6748	1.9506	0.9140	0.5127	2.5759	6.5338	6.6099	0.6971	0.5985	6.69
0.1770	0.2058	1.2932	0.8600	1.6851	1.9594	0.9141	0.5104	2.5864	6.6035	6.6788	0.6958	0.5984	6.67
0.1780	0.2067	1.2985	0.8613	1.6954	1.9683	0.9142	0.5080	2.5969	6.6740	6.7485	0.6946	0.5982	6.65
0.1790	0.2075	1.3037	0.8627	1.7057	1.9773	0.9143	0.5058	2.6074	6.7453	6.8190	0.6933	0.5981	6.63
0.1800	0.2083	1.3090	0.8640	1.7161	1.9862	0.9145	0.5035	2.6180	6.8173	6.8903	0.6920	0.5979	6.61
0.1810	0.2092	1.3142	0.8653	1.7266	1.9953	0.9146	0.5012	2.6285	6.8901	6.9623	0.6907	0.5977	6.59
0.1820	0.2100	1.3195	0.8667	1.7371	2.0044	0.9147	0.4989	2.6390	6.9637	7.0351	0.6895	0.5975	6.57
0.1830	0.2108	1.3247	0.8680	1.7477	2.0136	0.9149	0.4966	2.6495	7.0381	7.1087	0.6882	0.5974	6.55
0.1840	0.2117	1.3300	0.8693	1.7583	2.0228	0.9150	0.4944	2.6600	7.1132	7.1832	0.6870	0.5972	6.53
0.1850	0.2125	1.3353	0.8705	1.7690	2.0321	0.9152	0.4921	2.6705	7.1892	7.2584	0.6857	0.5970	6.51
0.1860	0.2134	1.3405	0.8718	1.7797	2.0414	0.9154	0.4899	2.6811	7.2660	7.3345	0.6845	0.5967	6.49
0.1870	0.2142	1.3458	0.8731	1.7905	2.0508	0.9155	0.4876	2.6916	7.3437	7.4114	0.6833	0.5965	6.47
0.1880	0.2150	1.3511	0.8743	1.8013	2.0603	0.9157	0.4854	2.7021	7.4222	7.4892	0.6820	0.5963	6.46
0.1890	0.2159	1.3563	0.8755	1.8122	2.0698	0.9159	0.4831	2.7127	7.5015	7.5679	0.6808	0.5961	6.44
0.1900	0.2167	1.3616	0.8768	1.8231	2.0794	0.9161	0.4809	2.7232	7.5817	7.6474	0.6796	0.5958	6.42

Then, from the table, (as $\frac{r}{L} = 2$, $\theta = 60^\circ$, $\beta = 30^\circ$) $\longrightarrow K_d = 0.28$

$$K_d = \frac{H_d}{H_i} = 0.28$$

$$\frac{H_d}{2.2} = 0.28$$

$$H_d = 0.28 \times 2.2$$

$$H_d = 0.62\text{m}$$

r/L	β (Degrees)												
	0	15	30	45	60	75	90	105	120	135	150	165	180
$\theta = 60^\circ$													
1/2	0.40	0.41	0.45	0.52	0.60	0.72	0.85	1.13	1.04	1.06	1.03	1.01	1.00
1	0.31	0.32	0.36	0.44	0.57	0.75	0.96	1.08	1.06	0.98	0.98	1.01	1.00
2	0.22	0.23	0.28	0.37	0.55	0.83	1.08	1.04	0.96	1.03	0.98	1.01	1.00
5	0.14	0.15	0.18	0.28	0.53	1.01	1.04	1.05	1.03	0.99	0.99	1.00	1.00
10	0.10	0.11	0.13	0.21	0.52	1.14	1.07	0.96	0.98	1.01	1.00	1.00	1.00
$\theta = 75^\circ$													
1/2	0.34	0.35	0.38	0.42	0.50	0.59	0.71	0.85	0.97	1.04	1.05	1.02	1.00
1	0.25	0.26	0.29	0.34	0.43	0.56	0.75	0.95	1.02	1.06	0.98	0.98	1.00
2	0.18	0.19	0.22	0.26	0.36	0.54	0.83	1.09	1.04	0.96	1.03	0.99	1.00
5	0.12	0.12	0.13	0.17	0.27	0.52	1.01	1.04	1.05	1.03	0.99	0.99	1.00
10	0.08	0.08	0.10	0.13	0.20	0.52	1.14	1.07	0.96	0.98	1.01	1.00	1.00
$\theta = 90^\circ$													
1/2	0.31	0.31	0.33	0.36	0.41	0.49	0.59	0.71	0.85	0.96	1.03	1.03	1.00
1	0.22	0.23	0.24	0.28	0.33	0.42	0.56	0.75	0.96	1.07	1.05	0.99	1.00
2	0.16	0.16	0.18	0.20	0.26	0.35	0.54	0.69	1.08	1.04	0.96	1.02	1.00
5	0.10	0.10	0.11	0.13	0.16	0.27	0.53	1.01	1.04	1.05	1.02	0.99	1.00
10	0.07	0.07	0.08	0.09	0.13	0.20	0.52	1.14	1.07	0.96	0.99	1.01	1.00

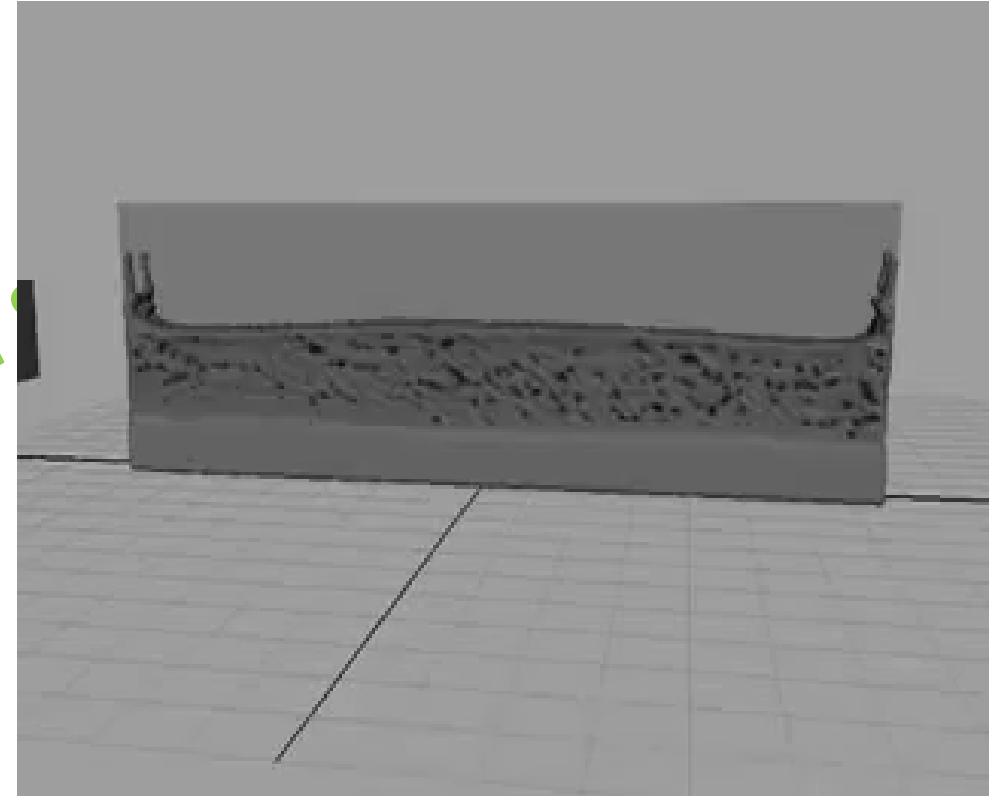
Dr. Mohar

Wave-Breaking

- Wave breaking is the process where a wave loses its stability and collapses, dissipating its energy.

Conditions for wave breaking:

- When horizontal particle velocity at the crest exceeds the celerity of the wave.
- When vertical particle acceleration is greater than the acceleration due to gravity.
- When the wave steepness, $H/L > 0.142 \tanh kd$.
- When the crest angle is less than 120° .
- When wave height is greater than $0.78d$.



- **Miche(1944)** describes wave breaking (regular wave) in a surf zone when the limiting wave steepness is:

$$\frac{H_b}{L_b} > 0.14 \tanh kd_b$$

- **Munk (1949)** derived several relationships from a modified **Solitary Wave Theory**:

Breaker Height Index

$$\frac{H_b}{H_0} = \frac{1}{3.3 \left(\frac{H_0}{L_0} \right)^{\frac{1}{3}}}$$

Breaker Depth Index

$$\frac{d_b}{H_b} = 1.28$$

- **Wave height at Breaking Point**

$$\frac{H_b}{H_0} = 0.5 \left(\frac{L_0}{H_0} \right)^{\frac{1}{5}}$$

Effect of bed slope on breaking process:

- Depending on the wave properties and the **angle of the bed slope**, the process of breaking takes place in various different ways.
- Battjes (1974) showed that the **Iribarren parameter**(ξ) guides this process.
- The Iribarren parameter (ξ) represents the ratio of the **slope steepness** $\tan \alpha$ and the **wave steepness**.

$$\xi = \frac{\tan \alpha}{\sqrt{H_0/L_0}}$$

Where, $\tan \alpha = m$ = steepness of the beach

H_0 = wave height in deep water

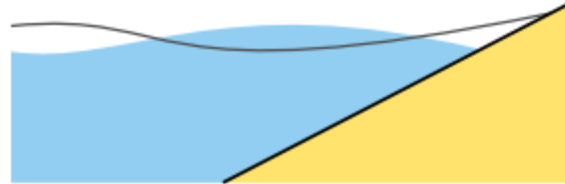
L_0 = wave length in deep water

- **Spilling Breakers:** $\xi < 0.5$
- **Plunging Breakers:** $0.5 < \xi < 3.3$
- **Surging Breakers:** $3.3 < \xi < 5$
- **Collapsing Breakers:** $\xi > 5$

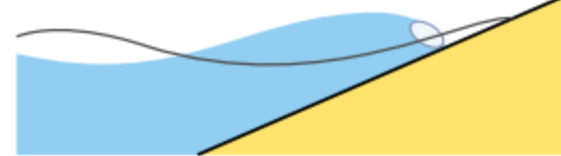
Types of wave breaking:

- **Spilling Breakers:** Waves break gradually, with foam spilling down the front. Common on gently sloping beaches.
- **Plunging Breakers:** Waves curl over and crash with significant energy. Often seen on steeper beaches.
- **Surging Breakers:** Waves don't break entirely but instead surge up the shore. Common on very steep beaches.
- **Collapsing Breakers:** A hybrid where the crest collapses, but there's no well-defined curl.

surging: $\xi > 5$



collapsing: $3.3 < \xi < 5$



plunging: $0.5 < \xi < 3.3$



spilling: $\xi < 0.5$



Problem

For a wave with period $T = 10$ s, deep water wave height $H_0 = 3.5$ m and beach slope $m = 1/15$. Calculate the breaking wave height using munk formulae . Also identify the type of breaking wave.

Solution, $T = 10\text{s}$, $H_0 = 3.5\text{m}$, $m = \frac{1}{15} = 0.067$

$$L_0 = 1.56 T^2 = 1.56 \times 10^2 = 156.16 \text{ m}$$

$$\frac{H_b}{H_0} = \frac{1}{3.3 \left(\frac{H_0}{L_0} \right)^{\frac{1}{3}}}$$

$$\frac{H_b}{3.5} = \frac{1}{3.3 \left(\frac{3.5}{156.16} \right)^{\frac{1}{3}}}$$

$$H_b = 2.675\text{m}$$

$$\xi = \frac{m}{\sqrt{H_0/L_0}}$$

$$\xi = \frac{0.067}{\sqrt{\frac{3.5}{56.16}}}$$

$$\xi = 0.268 < 0.5$$

Therefore, Spilling Breaker

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